

Preliminary Investigation of Activity Prediction in Nursing Homes using Activity History with Erroneous Time Stamps

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Abstract

Systems for recording nursing care activities using smartphones are becoming widespread in the nursing care field. As a result, it is becoming possible to predict future activities of care recipients based on the caregiving record history recorded in the system. Prediction of future activities will enable the development of various caregiving applications, such as preparation support for future caregiving activities, detection of missing entries in caregiving records, and sensor-based real-time activity recognition using the predictions as prior information. However, the caregivers are inherently busy and cannot record entries in real-time due to other tasks, resulting in many errors in time stamps of the entries, i.e., shift in occurrence time of an activity. When data with the many timing errors are used as training data in activity prediction methods, the performance of the prediction methods will deteriorate. In this paper, we propose an activity prediction method that is robust against the time errors. The proposed model has a module for correcting time errors and estimating whether or not a certain activity will occur from the current time to one hour later using the activity records of the past T hours. We evaluated the effectiveness of the proposed method using three datasets.

CCS Concepts

• **Human-centered computing** → *Empirical studies in ubiquitous and mobile computing*; • **Computing methodologies** → **Neural networks**; **Learning paradigms**.

Keywords

Nursing care, Activity prediction, Nursing care records

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1 Introduction

In recent years, the aging of the population in developed countries has increased the demand for nursing care. As a result, the efficiency of caregiving operations needs to be improved, and the digitization of caregiving records is attracting attention. Therefore, ubiquitous computing has been increasingly adopted in nursing care. An increasing number of nursing homes are introducing recording systems [1–3, 7] that allow care records to be entered using smartphones and tablet terminals. This system can not only record the body temperature, blood pressure, and other measurement data of care recipients but also activities performed by each care recipient, such as meal and excretion.

Figure 1 is an example of a binarized hourly care activity record, where “1” represents the time when the activity occurred and “0” represents the time when the activity did not occur. Using a recording system instead of paper records can reduce workload, and makes it easier to analyze the recorded data. For example, the short-term and long-term health status of a care recipient can be analyzed based on the data entered by the caregiver. The data accumulated in the recording system can also be used to predict activities that will occur in the future.

Many studies use caregiving records to predict future caregiving activity based on past activities [6, 9, 12]. If future caregiving activities can be predicted, caregivers can make necessary caregiving preparations and work more efficiently. Furthermore, the prediction of caregiving activities can be used as an aid in recording caregiving activities. The use of caregiving activity prediction can simplify the input of information into the recording system. For example, if care for a meal is predicted to occur within an hour, the recording system can reduce the time required for input by automatically filling input forms based on the predicted information. In addition, if a predicted caregiving activity has not yet been entered, the system

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activity	10:00~ 11:00	11:00~ 12:00	12:00~ 13:00	...
sleep	0	0	1	...
cook	0	1	0	...
⋮	⋮	⋮	⋮	...

care activity record

Figure 1: An example of binarized hourly care activity records. “1” shows an entry of an activity occurring at the corresponding time slot.

notifies the caregiver whether or not the caregiver has forgotten to record the caregiving activity, thereby preventing missing entries. Furthermore, the prediction of caregiving activities can be used to recognize caregiving activities. In other words, the prediction results using past caregiving activities can be used as prior knowledge in real-time caregiving activity recognition using caregiver sensor data. As described above, information from caregiver records is useful for predicting caregiving activities.

However, because the caregivers are inherently busy and cannot record entries in real-time due to other tasks, resulting in many errors in time stamps of the records. When records with many erroneous time stamps are used to train an activity prediction method, its prediction performance deteriorates.

In this paper, we propose a method for predicting future caregiving activity using nursing care records including time errors. We estimate whether or not a certain caregiving activity will occur from the current time to one hour later using the caregiving activity records for the past T hours. To deal with the problem of time errors, the proposed method is designed to have a module for correcting the time errors and leverage the outputs of the module for the prediction, enabling to predict future activities based on more accurate activity histories. The contributions of this work are as follows:

- We proposed a robust activity prediction method based on caregiving records with time errors as training data.
- We introduced the module that corrects time errors in the proposed model.
- We evaluated the effectiveness of the proposed method using 3 types of datasets.

2 Related Work

There are many studies on the prediction of daily activities. Sasaki et al. [11] proposed a method for predicting when activities will occur using LSTM based on data obtained from a smart home. Khaled et al. [4] proposed a unified deep learning model for monitoring elderly in execution of daily life activities. The proposed approach consists of three stages, which are activity recognition, anomaly detection, and next activity prediction. These studies used sensor data and other data obtained from smart homes. In this study, we use care records, which are obtained without installing specific equipment to predict activities.

If important future caregiving activities are predicted, caregivers can prepare for them. For this reason, research on predicting the activities of caregivers and care recipients has been active. Uchimura

et al. [12] proposed a model that uses caregiving records to predict urination time and urine volume in the bladder based on fluid intake. Prashant [5] developed a wearable device that measures urine volume in the bladder using NIR light. Although these studies focused on a specific activity, i.e., urination, the method in this study is designed to predict general caregiving activities.

There are studies on predicting caregiving activities using actual caregiving records collected at nursing homes. Saito et al. [9] focused on the error between the actual excretion caregiving time and the time on the caregiving record, and proposed a method for predicting the occurrence of excretion caregiving using machine learning such as random forest and support vector machines. Matsuki et al. [8] analyzed caregiving records and sensors to predict the activity level during the day based on caregivers’ sleep status. Inoue et al. [6] proposed a method to automatically generate caregiving records by recognizing caregivers’ activities using smartphone sensor data, and to predict the activities that caregivers will take the next day and their duration from the generated caregiving records. In contrast, our study focuses on activity prediction based on erroneous training data.

3 Proposed Method

3.1 Overview

An hourly occurrence of an activity is represented as a binary value, with 1 meaning the time when the activity occurred and 0 meaning the time when it did not occur. This binarized time series data is referred to as a binary occurrence time series, which is used as input of our method. Figure 1 is an example of a binary occurrence time series.

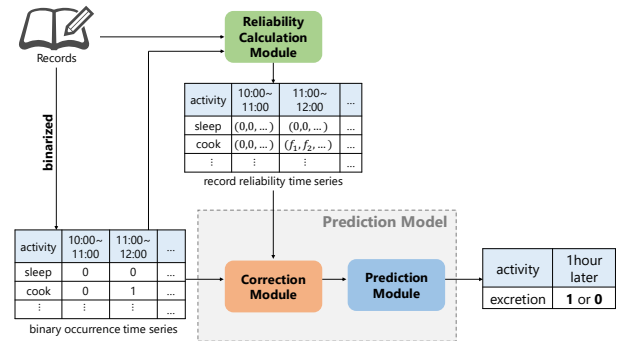


Figure 2: The overview of the proposed prediction model

Figure 2 shows an overview of the proposed method. The proposed method uses past activity records to predict whether a target activity will occur from the current time to 1 hour later. The proposed method mainly consists of the reliability calculation module and prediction model. The proposed prediction model is designed to have a correction module that corrects the time errors of records and a prediction module that predicts an activity occurrence of care recipients using the corrected records. We first create a binary occurrence time series using activity records for past T hours and use the created binary time series as input to the reliability calculation module. The reliability calculation module generates some features that reflect the reliability of the records. The correction module

uses the created features to correct the time errors in the binary occurrence time series. The corrected binary occurrence time series are then fed into the prediction module to predict whether or not the target activity will occur within 1 hour. The prediction model was trained as a whole.

3.2 Reliability Calculation Module

The reliability calculation module calculates some features for each activity occurrence using the record and auxiliary information, e.g., properties of a caregiver who input the entry (occurrence). The features are designed to reflect the reliability of the entry, and depend on datasets. The followings are examples of features.

- **Activity duration features:** The duration of the activity from the beginning to the end. Too long or too short activities can have time errors.
- **Activity correlation features:** Correlation coefficients between each pair of subjects are calculated for the activity class using a binary occurrence time series of the activity. The difference between the average of all the calculated correlation coefficients and the average correlation coefficients between a subject of interest and the others is used as a feature, showing the deviation from the others.
- **Subject features:** Features such as age and gender related to the caregivers themselves.

As shown in Figure 2, in output time series of this module, the calculated feature values were used as an element of a time slot corresponding to the original occurrence, while time slots when the activity did not occur were set to a zero vector. We call the resulting time series the record reliability time series.

3.3 Correction Module

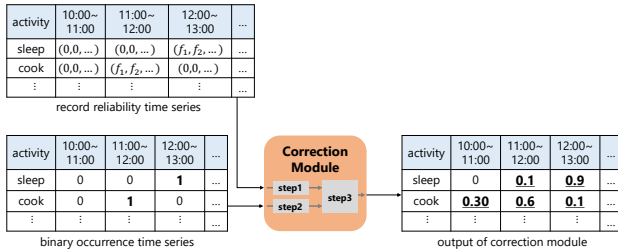


Figure 3: The overview of the correction module

Figure 3 shows the overview of the correction module. The inputs of the correction module are the binary occurrence time series and the record reliability time series. The correction module uses the record reliability time series to correct time errors in the input binary occurrence time series by distributing the value of 1 in the binary occurrence time series to the time slots before and after the original time slot.

We achieve this by preparing two trainable convolution kernels, each of them being responsible for shifting the occurrence time forward or backward. We then calculate the weights of the kernels to determine how to distribute the value of 1 by referring to the reliability features.

Figure 4 shows the overview of the procedure of the correction module. (Step 1) For each activity class, the correction module first calculates weights $\mathbf{W} = (\mathbf{W}_{no}, \mathbf{W}_f, \mathbf{W}_b)$, corresponding to weights of no-shift, forward-shift kernel, and backward-shift kernel, respectively, where W_{no}^t shows a weight value at time slot t . (Step 2) We use a CNN model with the kernels to process the binary occurrence time series for each activity class. The trainable kernels have the kernel size of 3 and initial values as shown in Figure 4. The sum of the values for the kernel is maintained as 1.

The CNN layer with the forward-shift kernel outputs $\mathbf{X}_f$, and the CNN layer with the backward-shift kernel outputs $\mathbf{X}_b$. The binary time series for the activity class corresponding to “no shift” is $\mathbf{X}_{no}$. (Step 3) Finally, according to the calculated weights $\mathbf{W}$, we aggregate $\mathbf{X}_{no}$, $\mathbf{X}_b$, and $\mathbf{X}_f$ to obtain the output of the correction module.

3.4 Prediction Module

Figure 5 shows the overview of the prediction module. The input of the prediction module is the output of the correction module. The prediction module outputs whether the target activity will occur within one hour after.

In general, the number of Class 1 instances representing activity occurrences in training data is smaller than the number of Class 0 instances. To address this problem of unbalanced data, we use undersampling during training to equalize the number of training data among the classes.

4 Experiments

4.1 Datasets

We used the following three datasets in the experiment. In addition to nursing record data, we also used activity datasets of diary study, which potentially contain entries with time stamp errors.

- Care record data obtained from an actual nursing home. For the training data, care records of 34 subjects from the period from May 1, 2018 to July 1, 2018, were used, and for the test data, care records of 29 subjects from the period from April 10, 2023 to April 14, 2023, were used. The ground truth labels for the test data were created by directly observing the caregivers’ activities by our research team. The dataset includes 11 types of activity classes. Of these, three activity classes, “Meal and medicine,” “Excretion,” and “Daytime user care,” which could be recorded by direct observation, were selected as the target activities for prediction. The input to the prediction model was the binary occurrence time series over 72 hours.
- Data_WP4_3_TAD [10] is a dataset about daily activities collected in a questionnaire format. The dataset includes 8 types of physical activities, 11 types of location related contexts, and 4 types of household situation information for 699 subjects, and they are used as input to the prediction model. There were 10 target activities and location related contexts for prediction including “Sleep” activity and “Indoor” location related context. The input to the prediction model was the binary occurrence time series over 72 hours.
- The American Time Use Survey (ATUS) [13] is a survey on time use conducted by the U.S. government. Subjects are

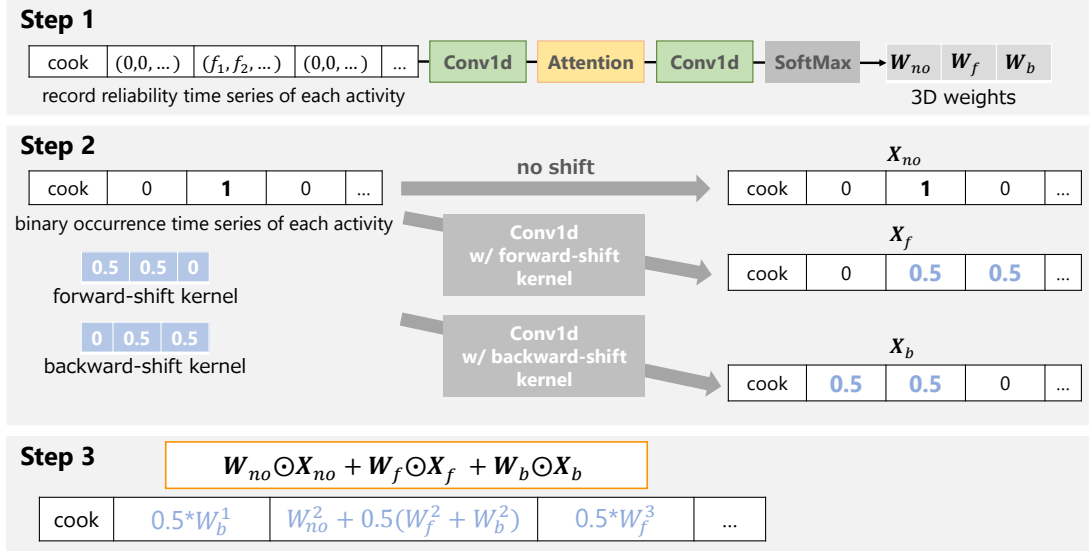


Figure 4: The overview of the procedure of the correction module

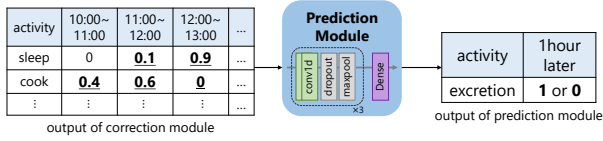


Figure 5: The overview of the prediction module

interviewed about how they spent the previous day, where they were, and with whom they were. In addition to how they spent the previous day, ATUS also records information such as gender, age, and occupation. In this paper, 16071 subjects with more than 30 occurrences of activities within 24 hours were selected for evaluation. The dataset includes 19 types of activities, of which 9 types were targeted for prediction. The input to the prediction model was the binary occurrence time series over 12 hours.

The record reliability time series used for the correction module varies depending on the dataset. For example, in the nursing care dataset, we used 3 types of features such as activity duration features and activity correlation features. In the TAD dataset, we used 10 types of features such as activity correlation features. In the ATUS dataset, we used 24 types of features such as activity duration features, activity correlation features, and subjects features.

4.2 Evaluation Procedures

We trained the model for 200 epochs and the model with the highest validation accuracy was used for testing. We used an Adam optimizer and a batch size of 32. The learning rate was 0.001.

The following methods were used for comparison.

- Baseline: The predicted value is determined as an entry at the same hour of the previous day. However, this method cannot be applicable to the ATUS dataset because this dataset has only one- data day per subject.

Table 1: The result of the experiment

	metrics	Nursing care	TAD	ATUS
Baseline	precision	0.289	0.735	-
	recall	0.299	0.795	-
	F1	0.294	0.764	-
	macroF1	0.563	0.862	-
	accuracy	0.728	0.931	-
w/o Correction	precision	0.426	0.634	0.119
	recall	0.522	0.914	0.859
	F1	0.469	0.749	0.208
	macroF1	0.664	0.848	0.512
	accuracy	0.776	0.914	0.701
Proposed Method	precision	0.460	0.876	0.160
	recall	0.535	0.746	0.832
	F1	0.495	0.806	0.269
	macroF1	0.682	0.889	0.573
	accuracy	0.793	0.950	0.793

- w/o Correction: It uses only the prediction module and does not use the correction module.

As evaluation metrics, we use the precision, recall, and F1 score for Class 1 corresponding to the occurrence of activities, as well as the macro F1 score and accuracy.

4.3 Results

Table 1 shows the results of the experiment. The proposed method recorded the highest macro F1 scores and accuracy for the all datasets. In particular, the proposed method showed an improvement in precision. We believe this is because the correction for the time errors helped capture the trends of the actual record and reduce the number of incorrect predictions.

Among the three types of activities in the nursing care dataset, the macro F1 score of the proposed method was higher than that

Table 2: The mean absolute error and correlation coefficient between the ground truth and the care records, and the mean absolute error and correlation coefficient between the ground truth and the output of the correction module.

	MAE	correlation coefficient
care records	0.196	0.260
output of the correction module	0.268	0.243

Table 3: The results of replacing the Conv1d in the prediction module with LSTM or RNN for the nursing care records dataset.

	metrics	CNN	LSTM	RNN
w/o Correction	precision	0.426	0.224	0.536
	recall	0.522	0.356	0.638
	F1	0.469	0.275	0.582
	macroF1	0.664	0.520	0.737
	accuracy	0.776	0.645	0.827
Proposed Method	precision	0.460	0.414	0.420
	recall	0.535	0.637	0.638
	F1	0.495	0.502	0.582
	macroF1	0.682	0.672	0.678
	accuracy	0.793	0.761	0.764

of Baseline for the “Daytime user care” and “Meal and medicine” activities. These activities are more periodic than the “Excretion” activity. This is because the proposed method is able to precisely capture the periodicity by correcting the time errors. Therefore, we believe that the proposed method is effective for activities with high periodicity.

4.4 Evaluation of Correction Module

To evaluate the correction module, we compared the output of the correction module and the care records recorded by caregivers using the nursing care dataset. Table 2 shows the mean absolute error and correlation coefficient between the ground truth obtained by direct observation and the care records recorded by caregivers. The table also shows the mean absolute error and correlation coefficient between the ground truth and the output of the correction module.

The result shows that the care records recorded by caregivers are closer to the ground truth than the output of the correction module. This is because even when there is no time error, the correction module distributes the value of 1 in the binary occurrence time series to the time slots before and after the original time slot. However, we believe that this behavior is effective when time errors are available.

4.5 Comparison with CNN and Recurrent Models

Table 3 shows the results for the nursing care dataset when we used LSTM or RNN in the prediction module instead of CNN. Contrary to our expectations, the macro F1 score of w/o Correction using RNN was the highest, and it was higher than that of the

proposed method using RNN. Furthermore, the macro F1 score of w/o Correction using RNN was significantly higher than that using CNN or LSTM. We believe that this is because RNN is good at handling short sequences of data and modeling simple time-series data, which is suitable for nursing care dataset. We believe that the proposed method using RNN can be improved by modifying the record reliability time series.

5 Conclusion

With the spread of ubiquitous computing in nursing, many studies focus on predicting nursing care activity using electronic records. However, caregivers often cannot record entries in real-time due to other tasks, leading to timestamp errors. We proposed reliability calculation and correction modules to address these errors in caregiving activity prediction. We evaluated their effectiveness using three datasets and confirmed its success on all of them. In the future, we plan to improve the accuracy by refining the features in the reliability calculation module.

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